A Data-Driven Analysis of Study Habits, Sleep Patterns, and Academic Pressure Among Students

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ABSTRACT-

This study looks at how study habits, sleep patterns, and academic pressure are connected for students. We used Google Forms to collect information from 17 students who filled out a survey. What we found out is that a lot of students feel okay to high academic pressure. When we looked closer at the numbers, we saw that the time students spend studying on weekdays does not really affect how much academic pressure they feel; it is actually a little bit the other way around. Students who study more on weekdays feel a bit less academic pressure. These results tell us that there are probably things that affect how stressed students are, it is not just about how much time they spend studying. The study habits and academic pressure of students are what we are really trying to understand, how sleep patterns, study habits, and academic pressure all work together for students.

INTRODUCTION-

In todays world students create a lot of data through their activities like how they study, when they sleep and how much time they spend on screens. With the help of Artificial Intelligence and data science we can look at these patterns to see what affects how well students do and how they feel. Many students feel pressure to do well in school and things like when they sleep how long they. How they use technology can impact how well they think and how stressed they are.

From a neuroscience point of view sleep and stress are closely linked to how our brains work how well we. How we handle emotions. By analyzing data we can learn more about how student behaviors relate to feeling stressed about school. This study looks at data from surveys to find patterns, between how students study, their sleep habits and how stressed they feel about school. The goal is to use data to see if the time students spend studying really affects how stressed they are and how well they are doing overall.

Research Questions

1. What are students' weekday and weekend study patterns?

2. What level of academic pressure do students experience?

3. Is there a measurable relationship between study hours and academic pressure?

4. How do sleep habits relate to students' daily routines and stress levels?

Methodology

Data for this study came from a questionnaire made with Google Forms.

The survey had questions, about students study habits.

It also covered their sleep schedules and lifestyle behaviors.

Additionally it looked at how academic pressure they felt.

The form was shared online.

Responses were collected without asking for names to help people answer honestly.

I collected responses from 37 students. Put them into Google Sheets. Then I cleaned up the student responses. Sorted them into groups. I made some charts and graphs to see if there were any patterns in how students study and the pressure they feel to do well in school. I looked at the student responses to see if there is a connection between how hours students study on weekdays and how much pressure they feel to do well in school using a simple way of looking at the student responses and the Google Sheets data, similar, to what people do when they look at student responses and Google Sheets data to understand study habits and academic pressure of the students.

The analysis followed basic data science techniques, including data cleaning, categorization of responses, visualization using charts, and correlation analysis to identify relationships between variables. These methods were used to interpret behavioral patterns within the dataset.

Results

1. Academic Pressure Levels

The survey results showed that students have levels of academic pressure. A lot of students 47.1 percent said they feel academic pressure. Then there are the students who feel academic pressure, which is 35.3 percent. Some students, 17.6 percent said they feel academic pressure. This means that most students have to deal with pressure every day when they are doing their school work. Academic pressure is something that affects the life of students and academic pressure is a big part of their daily routines.

2. Study Habits and Patterns

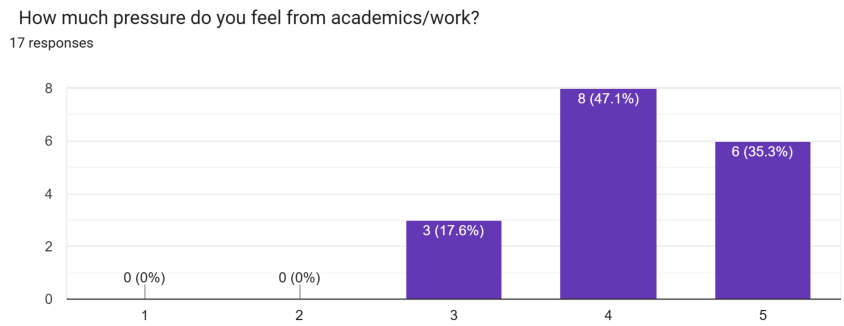

Figure 1 shows the

Figure 1 shows the distribution of perceived academic pressure among participants.

Students study for amounts of time on weekdays. Some study for a time while others study for a long time.

The study routines vary a lot.

This means that students have schedules and ways of managing their time.

Study habits are not the same for all students.

They study at times and, for different lengths.

3. Relationship Between Study Hours and Academic Pressure

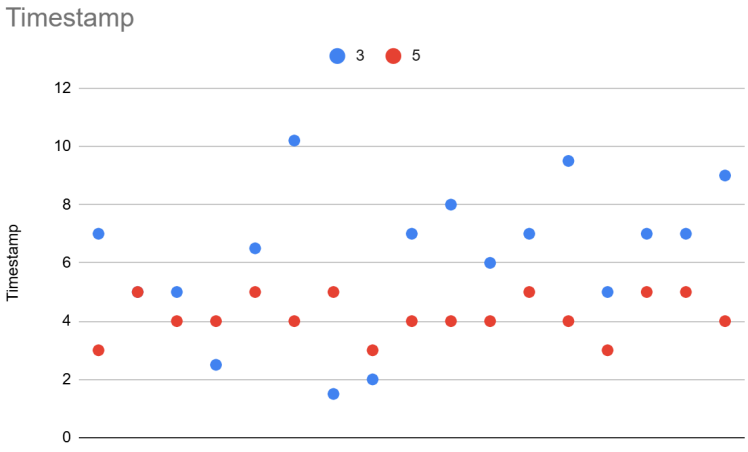

Figure 2 illustrates the relationship between weekday study hours and academic pressure.

I looked at the relationship between the hours I study on weekdays and how academic pressure I feel. I made a scatter plot. It seemed like the points were all over the place. There was no line that showed how these two things are connected. I also did a correlation test. Found out that the

connection between study hours and academic pressure is pretty weak. The number I got was -0.29 which means that studying more is only a bit related, to feeling less pressure. It does not really tell us if someone will feel stressed or not. The Relationship Between Study Hours and Academic Pressure is not that simple. Study Hours do not strongly affect Academic Pressure.

DISCUSSION

The results of this study show that many students experience pressure. Most students said they feel moderate to stress levels. It is interesting that the amount of time students spend studying does not really affect how academic pressure they feel.

The study found that the amount of time students study does not have an impact on academic pressure. This means that academic stress is common among students.. It is not just because of how long they study.

There are things that can cause stress and fatigue. For example using screens much before sleep and not having a regular daily routine can be bad for students. We need to look at student behavior to understand why students feel stressed.

From a neuroscience perspective not sleeping and using devices too much can be bad for our bodies. It can affect how well we can pay attention and make decisions. Using devices before bed can make it hard to sleep. This can make us feel tired and stressed during the day.

Overall academic stress is an issue. It is not, about how much time students spend studying. Academic stress is influenced by the way students live their lives. The academic pressure students experience is affected by things not just one thing like study time.

DATA SCIENCE INTERPRETATION

From a data science point of view the dataset showed patterns that we could not see by looking at individual responses.

Using visualization techniques helped us find out how academic pressure levels are distributed.

We also did a correlation analysis. Found out that there is only a weak link between how long students study and how stressed they feel.

This tells us that student well-being is affected by different factors, not just one thing that we can measure.

These kinds of datasets, about how people behave can show us how data analytics can be used to understand how people make decisions and behave.

The dataset and analysis also highlight the complexity of student being and academic pressure.

Data science can help us see these complexities clearly.

CONCLUSION

This study looked at how study habits, sleep patterns and academic pressure're connected for students. It used a lot of data to find out what is going on. The results showed that most students feel a lot of pressure to do well in school. This pressure is something that students deal with every day.

The study found that how hours students study on weekdays does not really affect how stressed they feel. This means that just studying for a time does not make students feel more stressed. Instead the study found that academic pressure is affected by things, including how much time students spend looking at screens and if they have regular routines or not.

Study habits and academic pressure are things to understand. If we do research with more students and look closer at what they do every day we might learn more about how students can be happy and successful, in school. This could help us figure out how students learn best and how to help them do well.

LIMITATIONS

This study has some limitations.

The sample size was small with 37 participants.

This may affect how much we can apply the findings to others.

The data came from participants themselves which means their answers might be influenced by their views or mistakes in estimating things.

The study only looked at a variables.

It did not consider factors, like the academic environment, personal expectations and mental health conditions.

Future studies could use more varied groups for a more accurate analysis.

References

American Psychological Association. (2020). *Publication manual of the American Psychological Association* (7th ed.).

Walker, M. (2017). *Why We Sleep*. Scribner.

Skills Demonstrated

- Survey Design
- Data Collection
- Data Cleaning
- Data Visualization
- Correlation Analysis
- Behavioral Data Interpretation
- Research Writing